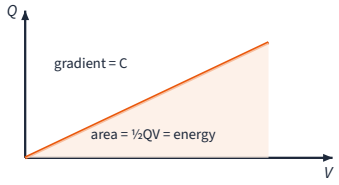
CHAPTER 13 CAPACITORS

Key idea: a capacitor stores charge and energy in the electric field between its plates.

1 CAPACITANCE



$$C = \frac{Q}{V}$$

- Unit farad: 1 F = 1 C V^{-1} (μF, nF, pF)
- Gradient of Q–V graph $= C$
- Isolated sphere: $C = 4\pi\varepsilon_0 R$

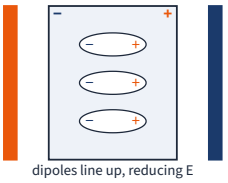
2 PARALLEL PLATES

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d}$$

$$E = \frac{V}{d} = \frac{\sigma}{\varepsilon_0}$$

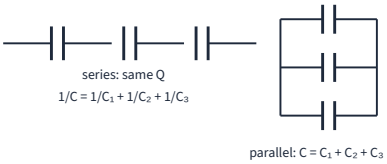
- **Battery connected:** V constant
- **Isolated:** Q constant

3 DIELECTRICS



Insert dielectric	Battery on	Isolated
C	$\times \varepsilon_r$	$\times \varepsilon_r$
Q	$\times \varepsilon_r$	same
V, E	same	$\div \varepsilon_r$
U	$\times \varepsilon_r$	$\div \varepsilon_r$

4 SERIES & PARALLEL



$$C_{\parallel} = C_1 + C_2$$

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

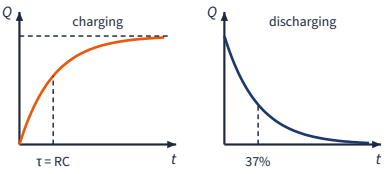
- Series: **same** Q; smaller C takes larger V
- Parallel: **same** V; charges add

5 ENERGY STORED

$$U = \tfrac{1}{2}QV = \tfrac{1}{2}CV^2 = \frac{Q^2}{2C}$$

- U = area under the V–Q graph
- Energy density $u = \tfrac{1}{2}\varepsilon_0 E^2$
- Battery gives QV: half stored, half lost as heat
- Sharing charge: Q conserved, energy lost

6 CHARGING & DISCHARGING



$$Q = Q_0 e^{-t/RC}$$

$$Q = Q_0\left(1 - e^{-t/RC}\right)$$

- $\tau = RC$: 37% left (discharge), 63% reached (charge)
- $t_{1/2} = RC \ln 2$
- Current decays as $I_0 e^{-t/RC}$ in **both** cases

7 UNIT CHECK

Quantity	Formula	Unit
Capacitance	Q/V	F
Permittivity	ε_0	F m^{-1}
Time constant	RC	s
Energy density	$\frac{1}{2}\varepsilon_0 E^2$	J m^{-3}

8 COMMON SECTION A TRAPS

- Using resistor rules for capacitors (they are reversed).
- Forgetting whether Q or V stays constant.
- μF → ×10^{-6} F; cm^2 → ×10^{-4} m^2.
- 63% is for charging, 37% for discharging.
- Ln graph of discharge: gradient $= -1/RC$.

EXAM TIPS

- ✓Decide first: battery connected or isolated?
- ✓For series, find the common Q first.
- ✓Use $\ln$ to find t from an exponential.
- ✓Quote τ before sketching charge–time graphs.